

ISTA 350 FINAL PROJECT

ADRIAN GONZALES

Gotta catch 'em all



THE DATASET



```
URL =  
"HTTPS://POKEMONDB.NET/POKEDEX/ALL"
```

```
HEADERS = {'USER-AGENT': "MOZILLA/5.0  
(WINDOWS NT 10.0; WIN64; X64)  
APPLEWEBKIT/537.36 (KHTML, LIKE GECKO)  
CHROME/91.0.4472.124 SAFARI/537.36"}
```

```
R = REQUESTS.GET(URL, HEADERS =  
HEADERS)  
TABLE = PD.READ_HTML(R.TEXT)  
POKEMON_DF = TABLE[0]
```



GLASS CANNON?



Code Used

```
#Glass cannon
def glass_cannon(df):
    """
    Creates a scatter plot w linear regression showing
    correlation between attack + speed on HP + defense +
    sp. defense

    | PARAMETER:
      df: pokemon dataframe
    """
    #Calculates pokemon power
    x = df["Attack"] + df["Speed"]
    #Calculates pokemon durability
    y = df["HP"] + df["Defense"] + df["Sp. Def"]

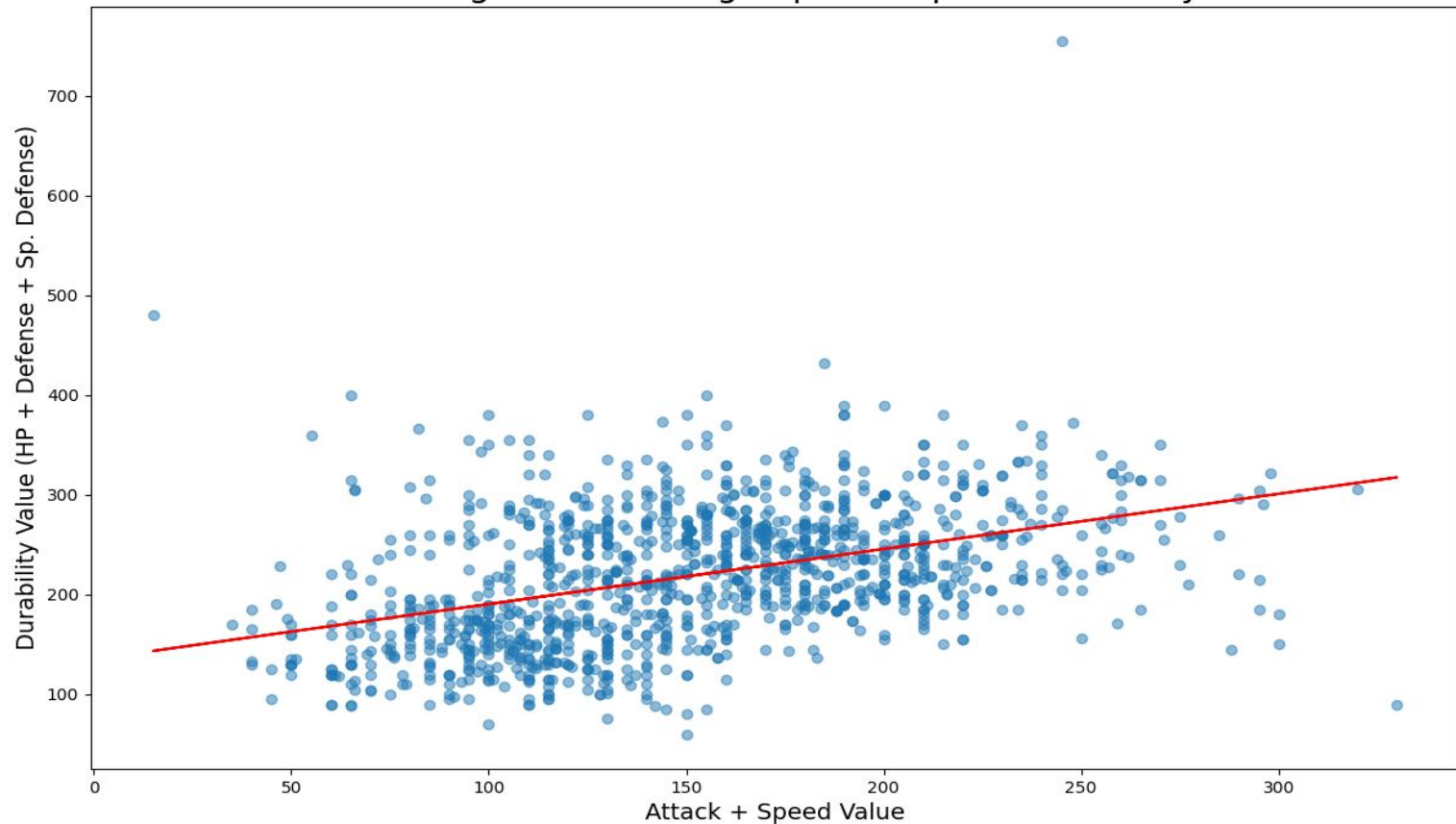
    #Used to find linear regression line
    x_const = sm.add_constant(x)
    model = sm.OLS(y, x_const)
    results = model.fit()
    slope = results.params.iloc[1]
    intercept = results.params.iloc[0]
    predicted = slope * x + intercept

    plt.figure()
    plt.scatter(x, y, alpha = 0.5)
    plt.plot(x, predicted, color = "red")
    plt.title("How High Attack + High Speed Impacts Durability", fontsize = 20)
    plt.xlabel("Attack + Speed Value", fontsize = 14)
    plt.ylabel("Durability Value (HP + Defense + Sp. Defense)", fontsize = 14)

    plt.show()
```


Relationship Between High Attack + Speed on Durability

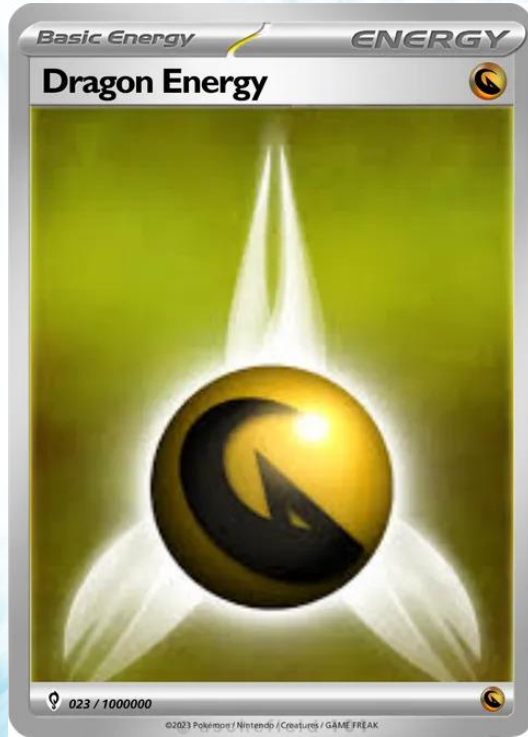
How High Attack + High Speed Impacts Durability



WHO'S THAT POKÉMON (TYPE)?



POKÉMON



POKÉMON

Code Used

```
#Best primary type
def primary_stats(df):
    """
    Creates a bar chart showing the average total stats
    for each Pokémon primary type.

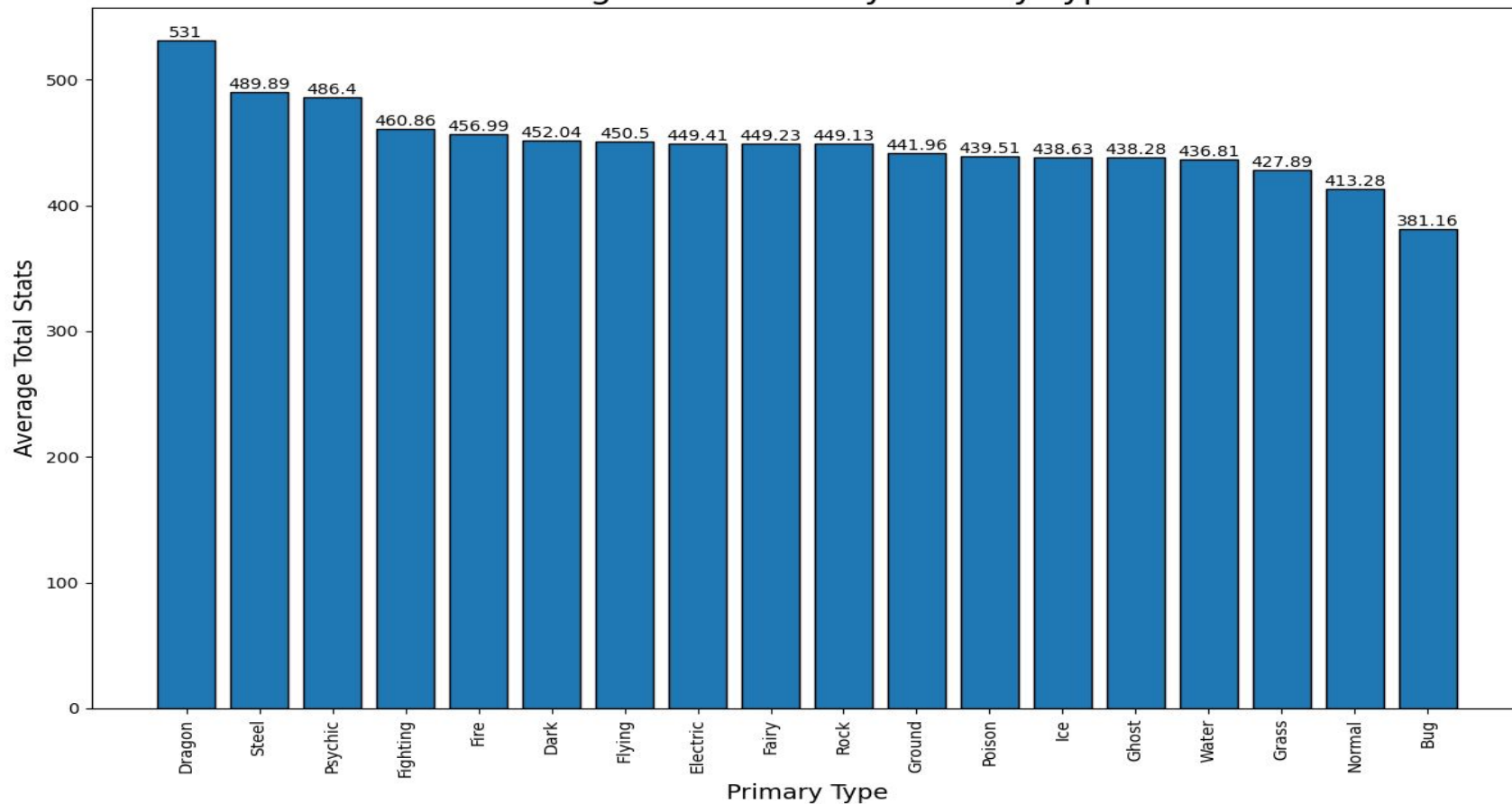
    PARAMETER:
        df: pokemon dataframe
    """
    df["Primary Type"] = df["Type"].str.split().str[0]
    mean_stat = df.groupby("Primary Type")["Total"].mean()
    mean_stat = mean_stat.sort_values(ascending = False)
    mean_stat = mean_stat.round(2)

    plt.figure()
    bars = plt.bar(mean_stat.index, mean_stat.values, edgecolor = "black")
    plt.bar_label(bars)
    plt.title("Average Total Stats by Primary Type", fontsize = 20)
    plt.xlabel("Primary Type", fontsize = 14)
    plt.ylabel("Average Total Stats", fontsize = 14)
    plt.xticks(rotation = 90)

    plt.show()
```


Pokémon's Strongest Primary Type

Average Total Stats by Primary Type



TOP 10 STRONGEST POKÉMON



Code Used

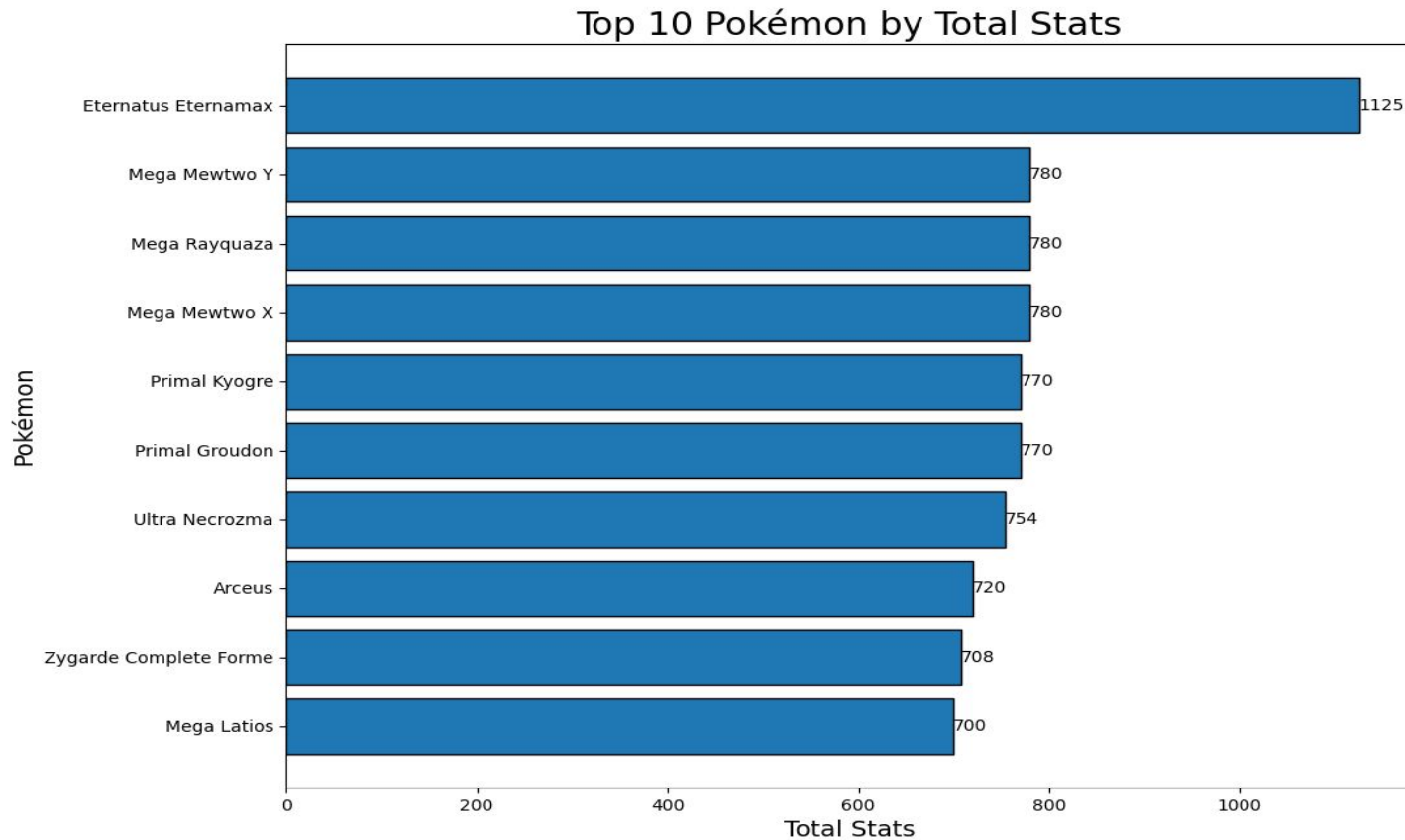
```
#Top 10 pokemon
def top_10_pokemon(df):
    """
    Creates a bar chart showing the top 10 strongest
    pokemon by total stat.

    PARAMETER:
        df: pokemon dataframe
    """
    top10 = df.sort_values("Total", ascending=False).head(10)

    plt.figure()
    bars = plt.barh(top10["Name"], top10["Total"], edgecolor="black")
    plt.bar_label(bars)
    plt.title("Top 10 Pokémon by Total Stats", fontsize=20)
    plt.xlabel("Total Stats", fontsize=14)
    plt.ylabel("Pokémon", fontsize=14)
    plt.gca().invert_yaxis()

    plt.tight_layout()
    plt.show()
```

Top 10 Pokémon



Works Cited

<https://pokemondb.net/pokedex/all>

<https://www.template.net/edit-online/375613/pokemon-environment-background>

<https://wall.alphacoders.com/big.php?i=723263>

<https://www.pinterest.com/pin/pokedex-template--122723158590936422/>

https://www.heraldscotland.com/news/14622702.Quiz_Who_s_that_Pokemon_/

<https://pokecardmaker.net/card/K0RA1D0N23/dragon-energy-ylj0yyh>

<https://collider.com/pokemon-powerful-ranked/>